

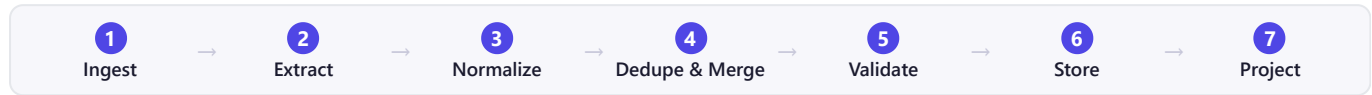
CandidateFusion

Design Doc

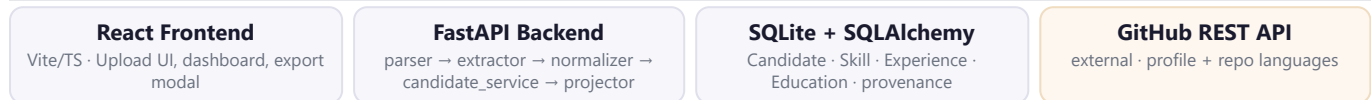
OBJECTIVE

Ingest candidate data from five heterogeneous sources (CSV, ATS JSON, resume PDF/DOCX, recruiter notes, GitHub URL), produce one deduplicated, normalized canonical record per candidate, and expose a runtime-configurable projection layer so any downstream consumer can reshape the output without backend changes.

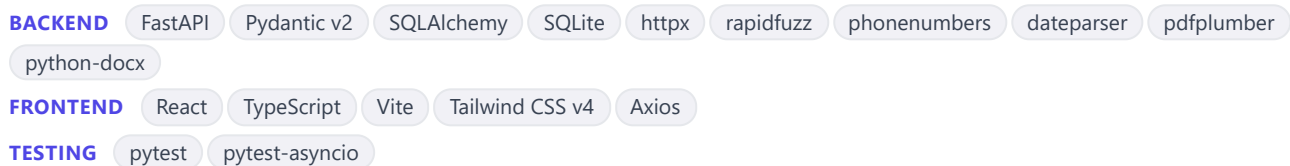
PIPELINE DIAGRAM



KEY COMPONENTS



TECH STACK



EXPECTED OUTPUT

A single canonical JSON record per candidate — normalized fields, per-field confidence, and source provenance — exportable as-is or reshaped at request time (selected fields, renamed paths, normalization, missing-value policy) via the `OutputConfig` projection.

THE REQUIRED TWIST: RUNTIME PROJECTION

A JSONPath-like resolver (`skills[].name` , `location.city`) lets a caller select, rename, and normalize fields, and choose how missing data is handled (`null` / `omit` / `error`) — all at request time.

One stored record, infinite output shapes — an ATS, a BI tool, and a recruiter export can each get a different JSON contract from the same data.

KEY DESIGN DECISIONS

- **Fuzzy dedup** (rapidfuzz, threshold 85) + email/phone exact-match catches real duplicates without false-merging.
- **Per-source confidence** — structured data (ATS 0.90) trusted more than free-text extraction (notes 0.55).
- **Skill confidence boosts** +0.05 per confirming source, capped at 1.0 — corroboration is signal.

SAMPLE OUTPUT (MERGED RECORD)

```
{
  "full_name": "Aisha Patel",
  "location": { "city": "San Francisco", "country": "US" },
  "skills": [{ "name": "Python", "confidence": 0.85,
               "sources": ["csv", "ats"] }],
  "fusion_score": 0.846
}
```

FUSION SCORE (CREATIVE ADDITION)

A single 0–1 quality signal beyond raw confidence — rewards *complete*, *trustworthy*, and *corroborated* records:

$$0.4 \times \text{completeness} + 0.4 \times \text{confidence} + 0.2 \times \text{cross-source verification}$$

ENGINEERING RIGOR

- 19 automated tests — pipeline integration, extractor, normalizer, projector, each independently unit-tested.
- Caught a hostname-spoofing bug in review: GitHub URL check used substring matching (`not-github.com` passed) — fixed with proper `urlparse` validation.

WHAT I'D DO WITH MORE TIME

Embedding-based similarity for higher-precision dedup at scale; Postgres instead of SQLite for concurrent writes; an async queue for large batch uploads; field-level audit history across re-ingests.